Stat4307 Final Exam Question 7 Cherry

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Import Cherry data set from CSV

```
cherry<-read.csv("C:\\Users\\super\\Downloads\\cherry.csv")
```

a.Plot volume vs. diameter of trees

```
plot(cherry$diameter, cherry$volume, data=cherry, xlab=("Tree Diameter"), ylab=("Tree Volume"), main=("
```

```
## Warning in plot.window(...): "data" is not a graphical parameter
```

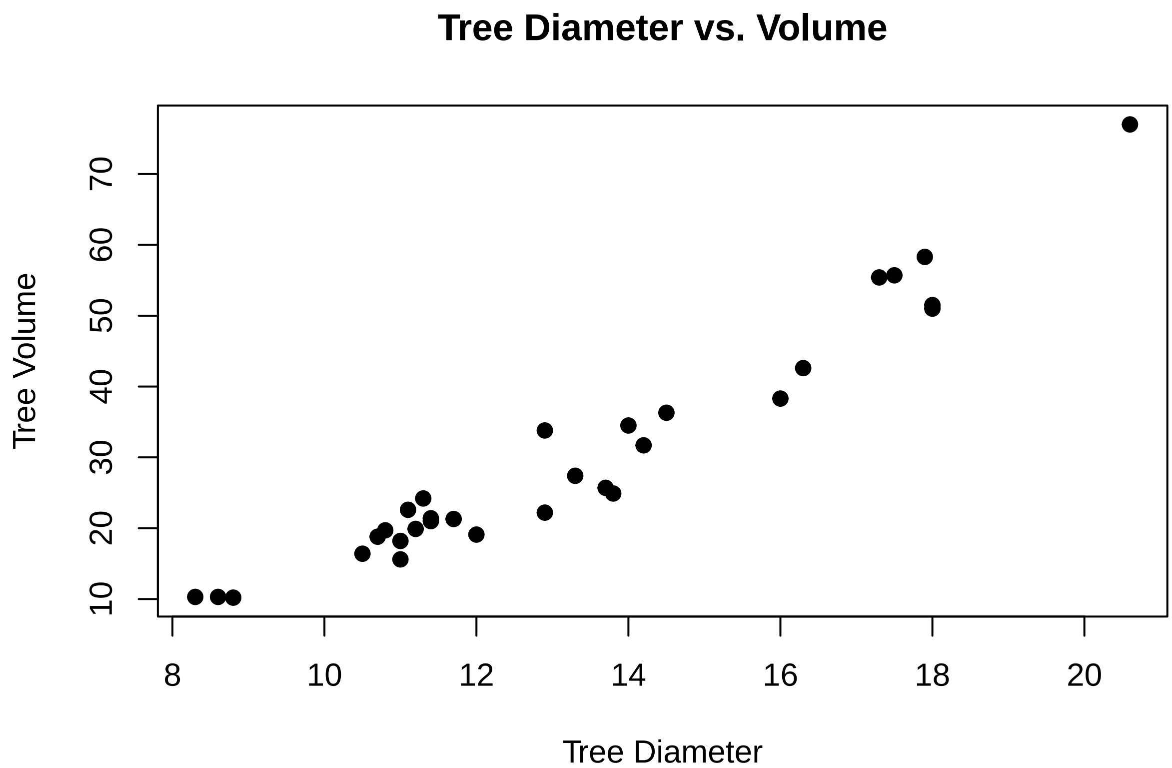
```
## Warning in plot.xy(xy, type, ...): "data" is not a graphical parameter
```

```
## Warning in axis(side = side, at = at, labels = labels, ...): "data" is not a
## graphical parameter
```

```
## Warning in axis(side = side, at = at, labels = labels, ...): "data" is not a
## graphical parameter
```

```
## Warning in box(...): "data" is not a graphical parameter
```

```
## Warning in title(...): "data" is not a graphical parameter
```



b.Suppose that these trees are an SRS from a forest of N=2967 trees and that the sum of the diameters for all trees in the forest is $t_x = 41,835in$. Use ratio estimation to estimate the total volume for all trees in the forest. Give a 95% confidence interval.

```
tx<-41835
n<-nrow(cherry)
N<-2967
vol.bar<-mean(cherry$volume)
tot.vol.est<-(tx/n)*N*vol.bar
se<-sqrt((tx^2/n^2)*((N-n)/(N-1))*var(cherry$volume))
lower<-tot.vol.est-qt(0.975,N-1)*se
upper<-tot.vol.est+qt(0.975,N-1)*se
cat("Estimated total volume of all Trees:", tot.vol.est, "cubic feet\n")
```

```
## Estimated total volume of all Trees: 120804988 cubic feet
```

```
cat("95% Confidence Interval:[",lower,",",upper,"] cubic feet\n")
```

```
## 95% Confidence Interval:[ 120761713 , 120848263 ] cubic feet
```

Here we can see that the estimated total volume for all trees in the forest is 120,804,988 cubic feet, with a 95% confidence interval of between 120,761,713 cubic feet and 120,848,263 cubic feet.

c. Use regression estimation to estimate the total volume for all trees in the forest. Give a 95% confidence interval.

```r
lin.mod <- lm(cherry$volume ~ cherry$diameter + cherry$height, data=cherry)
lin.mod
```

```
##
## Call:
## lm(formula = cherry$volume ~ cherry$diameter + cherry$height,
##     data = cherry)
##
## Coefficients:
##     (Intercept)  cherry$diameter    cherry$height
##        -57.9877           4.7082           0.3393
```